

Exponential and Logarithmic Equations



Exponential equations

- 1 Solve $3^{x+1} = 81$ exactly.
 - 2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.
 - 3 Solve $e^{2x} - 5e^x + 6 = 0$ exactly.
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Logarithmic equations

- 4 Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.
- 5 Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.
- 6 Solve $\log(x + 3) - \log(x - 1) = 1$, checking for extraneous solutions.
- 7 The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.